WORKED SOLUTION

STEP III 2016, Question 13

Probability and Statistics • Kurtosis of Sums

STEP III 2016, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q13

The key idea

Kurtosis is built only from deviations from the mean, so anything that shifts the variable leaves it unchanged; that lets us centre every variable at 0 and work with plain moments. For the Normal distribution, the fourth moment comes from integrating by parts, peeling off one factor of x to pair with the exponential. For a sum of independent mean-zero variables, expanding T^4 and taking expectations kills every term that contains some variable to the first power. This is the whole of part (ii), and it feeds straight into part (iii).

Solution

Shifting. Let $W = X - a$. By linearity of expectation, $\mathrm{E}(W) = \mu - a$, so

$$W - \mathrm{E}(W) = (X - a) - (\mu - a) = X - \mu.$$

The deviation of W from its mean is exactly the deviation of X from its mean. So W has the same variance $\mathrm{E}((X-\mu)^2) = \sigma^2$, the same fourth central moment $\mathrm{E}((X-\mu)^4)$, and hence the same kurtosis. In particular, taking $a = \mu$, every random variable has the same kurtosis as its centred version $X - \mu$, which has mean 0.

Part (i)

Let $X \sim \mathrm{N}(0, \sigma^2)$, with density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\,\mathrm{e}^{-x^2/(2\sigma^2)}, \qquad \text{so that} \qquad f'(x) = -\frac{x}{\sigma^2}\,f(x).$$

Since $\mu = 0$, we need $\mathrm{E}(X^4) = \int_{-\infty}^{\infty} x^4 f(x)\,\mathrm{d}x$.

Ask yourself How do we integrate $x^4\mathrm{e}^{-x^2/(2\sigma^2)}$ when $\mathrm{e}^{-x^2/(2\sigma^2)}$ has no elementary antiderivative?

The function $xf(x)$ *does* have an antiderivative, namely $-\sigma^2 f(x)$. So write $x^4 f(x) = x^3 \cdot xf(x)$ and integrate by parts with $u = x^3$ and $\mathrm{d}v = xf(x)\,\mathrm{d}x$, $v = -\sigma^2 f(x)$:

$$\int_{-\infty}^{\infty} x^4 f(x)\,\mathrm{d}x = \Big[-\sigma^2 x^3 f(x)\Big]_{-\infty}^{\infty} + 3\sigma^2 \int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x = 3\sigma^2 \int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x.$$

The boundary term vanishes because $x^3 \mathrm{e}^{-x^2/(2\sigma^2)} \to 0$ as $x \to \pm\infty$ (the exponential decays faster than any power grows). The same step with $u = x$ gives

$$\int_{-\infty}^{\infty} x^2 f(x)\,\mathrm{d}x = \Big[-\sigma^2 x f(x)\Big]_{-\infty}^{\infty} + \sigma^2 \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \sigma^2,$$

since f is a probability density and integrates to 1. (This confirms that the variance is σ^2.) Hence $\mathrm{E}(X^4) = 3\sigma^2 \cdot \sigma^2 = 3\sigma^4$, and

$$\kappa = \frac{3\sigma^4}{\sigma^4} - 3 = 0.$$

By the shifting result, every Normal distribution, whatever its mean, has kurtosis 0.

Part (ii)

Expand $T^4 = (Y_1 + \cdots + Y_n)^4$ by the multinomial theorem. Each term is a product of four factors Y_r, and the terms fall into five types according to how the four factors are shared among distinct indices:

$$Y_r^4, \qquad 4\,Y_r^3 Y_s, \qquad 6\,Y_r^2 Y_s^2, \qquad 12\,Y_r^2 Y_s Y_t, \qquad 24\,Y_r Y_s Y_t Y_u,$$

where the different letters stand for different indices. The coefficients count the arrangements: for example, $Y_r^2 Y_s^2$ arises from choosing which 2 of the 4 brackets supply Y_r, giving $\binom{4}{2} = 6$ ways. Each unordered pair $\{r, s\}$ appears once in the $Y_r^2 Y_s^2$ type.

Now take expectations. The Y's are independent, so the expectation of a product of powers of different Y's is the product of their expectations. In the types $Y_r^3 Y_s$, $Y_r^2 Y_s Y_t$ and $Y_r Y_s Y_t Y_u$, some variable appears to the first power only, and its factor $\mathrm{E}(Y_s) = 0$ makes the whole expectation zero. For example,

$$\mathrm{E}(Y_r^3 Y_s) = \mathrm{E}(Y_r^3)\,\mathrm{E}(Y_s) = 0.$$

Only the first and third types survive, and $\mathrm{E}(Y_r^2 Y_s^2) = \mathrm{E}(Y_r^2)\,\mathrm{E}(Y_s^2)$ for $r \neq s$, again by independence. Listing the pairs as $r < s$,

$$\mathrm{E}(T^4) = \sum_{r=1}^{n} \mathrm{E}(Y_r^4) + 6 \sum_{r=1}^{n-1} \sum_{s=r+1}^{n} \mathrm{E}(Y_r^2)\,\mathrm{E}(Y_s^2),$$

as required. (Note that this part used independence and zero means, but not that the Y's are identically distributed.)

Part (iii)

Let each X_r have mean μ and standard deviation σ, and let $S = X_1 + \cdots + X_n$. Put $Y_r = X_r - \mu$. The Y_r are independent and identically distributed with mean 0, and

$$\mathrm{E}(Y_r^2) = \sigma^2, \qquad \mathrm{E}(Y_r^4) = (\kappa + 3)\sigma^4,$$

the second by rearranging the definition of κ. Then $T = \sum Y_r = S - n\mu$. By the shifting result, S has the same kurtosis as T, so we work with T, which has mean 0.

By independence, $\operatorname{Var}(T) = \sum \operatorname{Var}(Y_r) = n\sigma^2$. The double sum in part (ii) has $\binom{n}{2} = \frac{1}{2}n(n-1)$ terms, each equal to σ^4, so

$$\begin{aligned} \mathrm{E}(T^4) &= n(\kappa+3)\sigma^4 + 6 \cdot \tfrac{1}{2}n(n-1)\sigma^4 \\ &= n\sigma^4(\kappa + 3 + 3(n-1)) = n\sigma^4(\kappa + 3n). \end{aligned}$$

Therefore the kurtosis of T, and so of S, is

$$\frac{\mathrm{E}(T^4)}{(\operatorname{Var} T)^2} - 3 = \frac{n\sigma^4(\kappa+3n)}{n^2\sigma^4} - 3 = \frac{\kappa}{n} + 3 - 3 = \frac{\kappa}{n},$$

as required.

Checks

- Take $Y = \pm 1$ with probability $\frac{1}{2}$ each. Then $\mathrm{E}(Y^4) = \sigma^4 = 1$, so $\kappa = 1 - 3 = -2$. For $n = 2$, the sum takes values $-2, 0, 2$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, so $\mathrm{E}(T^4) = 8$ and $\operatorname{Var} T = 2$. This gives kurtosis $8/4 - 3 = -1 = \kappa/2$, as predicted.
- With $n = 1$, part (iii) says nothing changes, as it must.
- Since a sum of independent Normals is Normal, part (iii) with $\kappa = 0$ must give 0 again, and it does.
- We confirmed by computer, using exact fractions, that the kurtosis of a sum of up to six independent copies of a three-valued variable is exactly κ/n. We also checked numerically that $\int x^4 f(x)\,\mathrm{d}x = 3\sigma^4$ for several values of σ.

Common slips

- Getting the coefficient of $Y_r^2 Y_s^2$ wrong: it is $\binom{4}{2} = 6$ per unordered pair (or 3 per ordered pair), not 4 or 12.
- Forgetting to centre the X_r before using part (ii), which needs mean 0.
- Dividing by σ^4 instead of $(n\sigma^2)^2$: the variance of the sum is $n\sigma^2$.

Beyond STEP

Kurtosis as defined here is the fourth *cumulant* divided by σ^4. Cumulants are the coefficients in the expansion of the logarithm of the moment generating function, and they simply add for independent sums. That is why the standardised fourth cumulant of a sum shrinks like $1/n$, and the third (skewness) like $1/\sqrt{n}$. This is a quantitative glimpse of the central limit theorem: the sum looks more and more Normal,

whose cumulants beyond the second are all zero. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions and the central limit theorem at this level.

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